Lecture 7: Quantum Phase Estimation

From Controlled-U to Eigenvalues – The Swiss Army Knife of Quantum Algorithms

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Lecture Overview 📋

1. **Quick Review** – Oracle model, controlled operations, Hadamard basis
2. **Controlled-U and Phase Kickback** – The fundamental building block
3. **Powers of Unitaries** – U, U^2, U^4, … and binary fractions
4. **Two-Qubit Example** – Controlling U and U^2 together
5. **Connection to QFT** – How binary numbers become phases
6. **Quantum Phase Estimation (QPE)** – The general algorithm
7. **Application: Finding Eigenvalues** – And why factoring is eigenvalue estimation
8. **Complexity and Circuit Implementation** – Costs and challenges
9. **Iterative Phase Estimation** – A qubit-efficient variant
10. **PennyLane Demos** – Code for phase estimation
11. **Exercises and Next Steps**

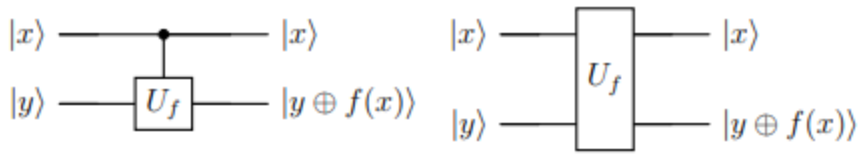
Quick Review – The Oracle Model and Controlled Operations 🔄

The Oracle as a Black Box

In quantum algorithms, we often assume access to a **unitary** U_f that implements a function f:

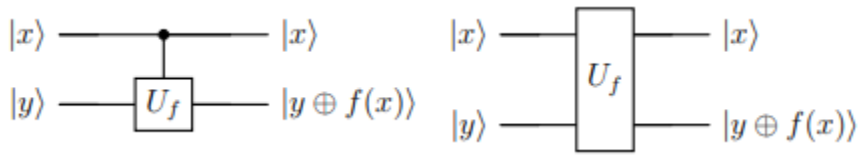
$$U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

This is the **standard oracle** we used in Deutsch–Jozsa, Bernstein–Vazirani, and Simon's algorithm.



Controlled Gates

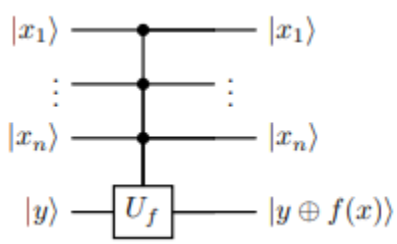
A **controlled-U** gate applies U to the target qubit(s) **only if** the control qubit is $|1\rangle$:



- If control = $|0\rangle$: target unchanged.
- If control = $|1\rangle$: target undergoes U.

The matrix (for a single control and U a 2×2 matrix) is:

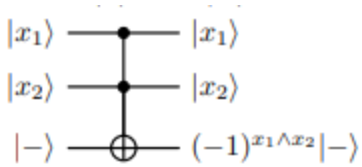
$$\text{controlled-}U = \begin{bmatrix} I & 0 \\ 0 & U \end{bmatrix}$$



Phase kick-back

When U changes phase of the target; this phase is **kicked back** to the control register.

When do we have phase-kick back?



The Hadamard Basis – $|+\rangle$ and $|-\rangle$

Recall:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

These states are the **eigenstates** of the Pauli-X gate:

$$X|+\rangle = +|+\rangle, \quad X|-\rangle = -|-\rangle$$

They are central to phase kickback.

Controlled-U and Phase Kickback 🦵

The Basic Idea

Suppose the target qubit is in an **eigenstate** of U with eigenvalue $e^{i\phi}$:

$$U|\psi\rangle = e^{i\phi}|\psi\rangle$$

Now consider a controlled-U with the control qubit in a superposition:

1. **Control =** $|0\rangle$**:** target unchanged.
2. **Control =** $|1\rangle$**:** target gets multiplied by $e^{i\phi}$.

Kickback with $|+\rangle$

Start with the control in $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and the target in the eigenstate $|\psi\rangle$:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |\psi\rangle$$

Apply controlled-U:

$$\frac{1}{\sqrt{2}}\big(|0\rangle|\psi\rangle + |1\rangle U|\psi\rangle\big) = \frac{1}{\sqrt{2}}\big(|0\rangle|\psi\rangle + e^{i\phi}|1\rangle|\psi\rangle\big)$$

Factor out $|\psi\rangle$:

$$= \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}} \otimes |\psi\rangle$$

Observation: The target state $|\psi\rangle$ is **unchanged**! The phase $e^{i\phi}$ has been **kicked back** onto the control qubit. 🎯

Kickback with $|-\rangle$

What if the control starts in $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$?

$$\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \otimes |\psi\rangle$$

After controlled-U:

$$\frac{1}{\sqrt{2}}\big(|0\rangle|\psi\rangle - e^{i\phi}|1\rangle|\psi\rangle\big) = \frac{|0\rangle - e^{i\phi}|1\rangle}{\sqrt{2}} \otimes |\psi\rangle$$

Again the target is unchanged, and a phase appears on the $|1\rangle$ component of the control.

The General Principle

> **Phase Kickback:** If a controlled-U acts on a control qubit and a target that is an eigenstate

> of U, the eigenvalue $e^{i\phi}$ appears as a relative phase on the $|1\rangle$ component of the control, while the target remains in the eigenstate.

This is the core mechanism behind **quantum phase estimation**.

Extracting the Phase via Hadamard

$$\frac{1}{\sqrt{2}}\left(|0\rangle|\psi\rangle + e^{i\phi}|1\rangle|\psi\rangle\right) = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}} \otimes |\psi\rangle$$

If we apply a **Hadamard gate** to the control qubit **after** the phase kickback, we can extract information about ϕ.

Let's test with control initially in $|0\rangle$:

1. $|0\rangle \xrightarrow{H} |+\rangle$
2. Controlled-U with eigenstate target: $|+\rangle \rightarrow \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}$
3. Apply H again:

$$H\left(\frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}\right) = \frac{1+e^{i\phi}}{2}|0\rangle + \frac{1-e^{i\phi}}{2}|1\rangle$$

The probability of measuring $|0\rangle$ is:

$$P(0) = \left|\frac{1+e^{i\phi}}{2}\right|^2 = \frac{1+cos\phi}{2} = \cos^2\left(\frac{\phi}{2}\right)$$

Similarly, $P(1) = \sin^2\left(\frac{\phi}{2}\right)$.

Result: The measurement probabilities depend on the unknown phase ϕ! This is the first glimpse of how we can estimate ϕ using quantum circuits.

The Key Mathematical Foundation: Functions of Matrices

functions of matrices are functions of eigenvalues:

$$f(U)|\psi\rangle = f(\lambda)|\psi\rangle$$

where λ is the eigenvalue for eigenvector $|\psi\rangle$.

The Key Observation

If $U|\psi\rangle = e^{i\phi}|\psi\rangle$, then for any integer k:

$$U^k|\psi\rangle = e^{ik\phi}|\psi\rangle$$

The eigenvalue gets raised to the k-th power: $(e^{i\phi})^k = e^{ik\phi}$.

Binary Representation of Phases

Suppose the phase ϕ is of the form:

$$\phi = 2\pi \times \theta, \qquad \theta \in [0, 1)$$

We can write θ as a **binary fraction**:

$$\theta = 0.\theta_1\theta_2\theta_3\ldots\theta_n = \sum_{j=1}^{n} \theta_j 2^{-j}, \quad \theta_j \in \{0, 1\}$$

For example, $\theta = 0.75 = 0.11_2$ means:

$$\theta = 1 \cdot 2^{-1} + 1 \cdot 2^{-2} = 0.5 + 0.25 = 0.75$$

The "Decimal Point Shifting" Trick

Consider $\theta = 0.\theta_1\theta_2\theta_3\ldots$

Multiply by 2 (the bits shift):

- $2\theta = \theta_1.\theta_2\theta_3\ldots$

Multiply by 2^k:

- $2^k\theta = \theta_1\theta_2\ldots\theta_k.\theta_{k+1}\ldots$

This shifting is exactly what happens when we use **powers of** U:

- U gives phase $e^{i2\pi\theta}$ (i.e., $2\pi \times 0.\theta_1\theta_2\theta_3\ldots$)
- U^2 gives phase $e^{i2\pi(2\theta)} = e^{i2\pi(\theta_1.\theta_2\theta_3\ldots)}$

$e^{i2\pi(\theta_1.\theta_2\theta_3\ldots)} = e^{i2\pi(\theta_1)} \times e^{i2\pi(0.\theta_2\theta_3\ldots)}$

- $e^{i2\pi\theta_1} = 1$ (since $\theta_1 \in \{0, 1\}$),

so the effective phase is $e^{i2\pi\times 0.\theta_2\theta_3\cdots}$.

- U^4 gives phase $e^{i2\pi(4\theta)} = e^{i2\pi(\theta_1\theta_2.\theta_3\ldots)}$
  The effective fractional part is $0.\theta_3\theta_4\ldots$.

Example: $\theta = 0.75 = 0.11_2$

$$\phi = 2\pi\theta, \quad \theta \in [0, 1)$$

Example:

$$\theta = 0.75_{10} = 0.11_2$$

Multiplying by 2 Shifts the Binary Point

$$0.11_2 \times 2 = 1.1_2$$

- Integer part 1 gives phase 2π (invisible)
- Fractional part 0.1_2 is the effective phase

Phase from U

$$U \longleftrightarrow \text{ multiply by } 2^0 = 1$$

$$\text{Fraction} = 0.11_2 = \frac{1}{2} + \frac{1}{4} = 0.75$$

$$\phi = 2\pi \times 0.75 = 1.5\pi \quad \Longrightarrow \quad e^{i1.5\pi} = -i$$

Phase from U^2

$$U^2 \longleftrightarrow \text{ multiply by } 2^1 = 2$$

$$0.11_2 \times 2 = 1.1_2 \longrightarrow \text{ effective fraction } 0.1_2$$

$$0.1_2 = \frac{1}{2} = 0.5$$

$$\phi = 2\pi \times 0.5 = \pi \quad \Longrightarrow \quad e^{i\pi} = -1$$

Phase from U^4

$$U^4 \longleftrightarrow \text{ multiply by } 2^2 = 4$$

$$0.11_2 \times 4 = 11.0_2 \longrightarrow \text{ effective fraction } 0.0_2$$

$$0.0_2 = 0$$

$$\phi = 2\pi \times 0 = 0 \quad \Longrightarrow \quad e^{i0} = 1$$

The General Pattern

For $\theta = 0.\theta_1\theta_2\theta_3\ldots_2$:

$$U^{2^j} \;\longleftrightarrow\; 2^j\theta = \theta_1 \cdots \theta_j \,.\, \theta_{j+1}\theta_{j+2}\ldots_2$$

$$e^{i2\pi(2^j\theta)} = e^{i2\pi(0.\theta_{j+1}\theta_{j+2}\ldots_2)}$$

Each power of U "shifts out" the most significant bit of θ.

Two-Qubit Example – Controlling U and U^2

two-qubit phase estimation

We use **two control qubits** and one target qubit prepared in an eigenstate $|\psi\rangle$ of U.

- **Top control qubit** (call it q_1) – controls U^2
- **Bottom control qubit** (call it q_0) – controls $U^1 = U$

We write the combined state with the **most significant qubit first**: $|q_1 q_0\rangle$.

- Target eigenstate $|\psi\rangle$ with $U|\psi\rangle = e^{i2\pi\theta}|\psi\rangle$, $\theta = 0.11_2$.

Gates

- q_0 controls U (multiplies phase by $1 \cdot \theta$).
- q_1 controls U^2 (multiplies phase by $2 \cdot \theta$).

What Happens to Each Control Qubit?

Start with both controls in $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$.

For q_0 (controls U):
When $q_0 = |1\rangle$, the state picks up phase $e^{i2\pi\theta}$.
So the state of q_0 becomes:

$$\frac{|0\rangle + e^{i2\pi\cdot 0.11_2}|1\rangle}{\sqrt{2}}$$

For q_1 (controls U^2):
When $q_1 = |1\rangle$, the state picks up phase $e^{i2\pi(2\theta)}$.
$2\theta = 0.11_2 \times 10_2 = 1.1_2$.
The integer part 1 gives $e^{i2\pi} = 1$, so the effective phase is $e^{i2\pi\cdot 0.1_2}$.

Thus the state of q_1 becomes:

$$\frac{|0\rangle + e^{i2\pi\cdot 0.1_2}|1\rangle}{\sqrt{2}}$$

The Full State is a Tensor Product

the **control register remains in a product state** with the target:

$$|\psi_{\text{controls}}\rangle \otimes |\psi\rangle$$

where

$$|\psi_{\text{controls}}\rangle = \left(\frac{|0\rangle + e^{i2\pi\cdot 0.1_2}|1\rangle}{\sqrt{2}}\right)_{q_1} \otimes \left(\frac{|0\rangle + e^{i2\pi\cdot 0.11_2}|1\rangle}{\sqrt{2}}\right)_{q_0}$$

The control-register is Exactly the QFT Tensor Product

Recall the factorised form of the QFT for $n = 2$ qubits. For an input state $|j_1 j_2\rangle$:

$$\mathrm{QFT}|j_1 j_2\rangle = \frac{1}{\sqrt{2}}\Big(|0\rangle + e^{2\pi i\, 0.j_2}|1\rangle\Big)_{q_0} \otimes \frac{1}{\sqrt{2}}\Big(|0\rangle + e^{2\pi i\, 0.j_1 j_2}|1\rangle\Big)_{q_1}$$

(Note: The usual QFT formula places the least significant qubit first in the tensor product; the order can be swapped with SWAPs. Here q_0 is LSB, q_1 is MSB.)

Compare with our state:

- q_0 has phase $e^{2\pi i\, 0.11_2}$ → matches $0.j_2$ with $j_2 = 1$.
- q_1 has phase $e^{2\pi i\, 0.1_2}$ → matches $0.j_1 j_2$ with $j_1 = 1$, $j_2 = 1$.

So our control register is exactly $\mathrm{QFT}|11\rangle$ (up to qubit ordering).

Extracting the Binary Digits

- Applying the **inverse QFT** to the control register ($|\psi_{\text{controls}}\rangle$) yields the computational basis state $|11\rangle$.
- A measurement then gives 11, revealing that $\theta = 0.11_2$.

Key takeaway:

- Each control qubit ends up in a state $|0\rangle + e^{i2\pi\cdot(\text{binary fraction})}|1\rangle$.
- The binary fractions are precisely those appearing in the QFT tensor product.
- Therefore, inverse QFT directly reads out the binary digits of θ.

Alternative View: Recall the Quantum Fourier Transform

In Lecture 6, we saw that the QFT maps a computational basis state $|k\rangle$ to:

$$\mathrm{QFT}|k\rangle = \frac{1}{\sqrt{N}}\sum_{j=0}^{N-1} e^{2\pi i\, jk/N}|j\rangle$$

For $N = 2^n$, the QFT encodes the integer k into **phases** of the superposition.

The Inverse Relationship

The state we produced in the two-qubit phase estimation:

$$\frac{1}{2}\sum_{x=0}^{3} e^{i2\pi(x\theta)}|x\rangle$$

Look familiar? If we write $\theta = k/4$ for some integer k (i.e., θ is a 2-bit binary fraction), then:

$$e^{i2\pi(x\theta)} = e^{2\pi i\, xk/4}$$

This is **exactly** the coefficient that appears in the **inverse QFT** of $|k\rangle$!

For Our Example $\theta = 0.11_2$

The state after phase kickback is:

$$\frac{1}{2}\Big(|00\rangle - i|01\rangle - |10\rangle + i|11\rangle\Big)$$

If we apply the **inverse QFT** (or, depending on convention, the QFT) to this state, we get back $|11\rangle$ – the binary representation of θ!

Quantum Phase Estimation (QPE) – The General Algorithm

Problem Statement

Given:

- A unitary operator U (as a controlled gate)
- An eigenstate $|\psi\rangle$ such that $U|\psi\rangle = e^{i2\pi\theta}|\psi\rangle$ with unknown $\theta \in [0, 1)$

Goal: Estimate θ to n bits of precision.

The Algorithm

1. **Initialization:**
    - First register: n qubits in $|0\rangle^{\otimes n}$
    - Second register: the eigenstate $|\psi\rangle$ (prepared somehow)
2. **Create superposition:**
   Apply $H^{\otimes n}$ to the first register:

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle|\psi\rangle$$

3. **Apply controlled-U^{2^j}:**
   For each qubit j (from 0 to $n-1$), apply controlled-U^{2^j} with the j-th qubit as control and $|\psi\rangle$ as target.
   This kicks back phases $e^{i2\pi(2^j\theta)}$:

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} e^{i2\pi\theta x}|x\rangle|\psi\rangle$$

4. **Apply inverse QFT:**
   Apply $\mathrm{QFT}^\dagger$ to the first register.
   This transforms the state to a peak around $|2^n\theta\rangle$ (rounded to nearest integer).
5. **Measure:**
   Measure the first register to obtain an integer $y \in \{0, \ldots, 2^n - 1\}$.
   The estimate is $\tilde{\theta} = y/2^n$.

Accuracy and Success Probability

- If θ can be exactly represented with n bits, the algorithm yields θ with **certainty**.
- If θ is not an exact binary fraction, the measurement gives the closest n-bit approximation with probability $\geq 4/\pi^2 \approx 0.405$.
- By adding extra qubits and repeating, we can boost success probability arbitrarily close to 1.

Application – Finding Eigenvalues (and Factoring)

Many quantum algorithms reduce to **eigenvalue estimation**:

- **Shor's factoring algorithm:** Find the order r of $a \bmod N$. The unitary $U|y\rangle = |ay \bmod N\rangle$ has eigenvalues $e^{i2\pi s/r}$ for $s = 0, \ldots, r-1$. The phase $\theta = s/r$ directly gives the period r via continued fractions.
- **Quantum chemistry:** Finding ground state energies of molecules (eigenvalues of the Hamiltonian).
- **Optimization:** Finding eigenvalues of certain matrices.

Shor's Algorithm Revisited

The modular exponentiation oracle U_f

In Lecture 6c we assumed a unitary U_f that computes the function $f(x) = a^x \bmod N$:

$$U_f\,|x\rangle|0\rangle = |x\rangle|a^x \bmod N\rangle.$$

This is the oracle used in the period-finding circuit: after applying U_f to a superposition over x, measuring the work register collapses the input register to a periodic comb $|x_0 + jr\rangle$.

Building U_f from controlled modular multiplications

Exponentiation by repeated squaring expresses a^x as a product of powers a^{2^j}:

$$a^x = \prod_{j=0}^{m-1} (a^{2^j})^{x_j}, \qquad x = \sum_{j=0}^{m-1} x_j 2^j,\ x_j \in \{0,1\}.$$

To compute this product into a work register, we need a **controlled modular multiplication** for each j:

$$\text{c-}U_{a^{2^j}} \ :\ |x_j\rangle|y\rangle \ \longmapsto\ |x_j\rangle|y \cdot (a^{2^j})^{x_j} \bmod N\rangle.$$

Define the basic modular multiplication unitary

$$U_a\,|y\rangle = |ay \bmod N\rangle.$$

Then $U_{a^{2^j}} = (U_a)^{2^j}$, so the controlled gates required are exactly

$$\text{controlled-}U_a,\ \text{controlled-}U_a^2,\ \text{controlled-}U_a^4,\ \ldots,\ \text{controlled-}U_a^{2^{m-1}}.$$

Thus the modular exponentiation circuit is **built from controlled powers of U_a**.

Phase estimation with U_a uses the same controlled powers

The Quantum Phase Estimation (QPE) algorithm requires applying controlled powers of a unitary to an eigenstate. If we choose the unitary to be U_a, the needed gates are precisely the ones listed above. Hence **the same hardware that computes $a^x \bmod N$ can be repurposed to run phase estimation on U_a**.

Eigenstates and eigenvalues of U_a

Because $a^r \equiv 1 \pmod{N}$, we have $U_a^r = I$ on the subspace spanned by $\{|a^k \bmod N\rangle\}$. The eigenvalues of U_a are therefore r-th roots of unity:

$$\lambda_s = e^{2\pi i\, s/r}, \qquad s = 0, 1, \ldots, r-1.$$

The corresponding eigenstates are

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i\, sk/r}\, |a^k \bmod N\rangle.$$

Verifying the Eigenvalue Equation for $|u_s\rangle$

We start with the definition of the eigenstate:

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i\, sk/r}\, |a^k \bmod N\rangle.$$

Apply the unitary U_a to $|u_s\rangle$:

$$U_a|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i\, sk/r}\, U_a|a^k \bmod N\rangle.$$

By definition, $U_a|y\rangle = |ay \bmod N\rangle$, so:

$$U_a|a^k \bmod N\rangle = |a \cdot a^k \bmod N\rangle = |a^{k+1} \bmod N\rangle.$$

Thus we have:

$$U_a|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i\, sk/r}\, |a^{k+1} \bmod N\rangle.$$

Eigenvalue Equation

$$
\begin{aligned}
U_a|u_s\rangle &= \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} \, |a^{k+1} \bmod N\rangle \\
&= \frac{1}{\sqrt{r}} \sum_{j=1}^{r} e^{-2\pi i s (j-1)/r} \, |a^{j} \bmod N\rangle \qquad (j = k+1) \\
&= \frac{1}{\sqrt{r}} \sum_{j=1}^{r} \left(e^{-2\pi i s j/r} \cdot e^{2\pi i s/r} \right) |a^{j} \bmod N\rangle \\
&= e^{2\pi i s/r} \cdot \frac{1}{\sqrt{r}} \sum_{j=1}^{r} e^{-2\pi i s j/r} \, |a^{j} \bmod N\rangle.
\end{aligned}
$$

Because $a^r \equiv 1 \pmod{N}$, the term $j = r$ is equal to $j = 0$: $|a^r \bmod N\rangle = |1\rangle = |a^0 \bmod N\rangle$, and $e^{-2\pi i s r/r} = e^{-2\pi i s} = 1$. Thus the sum over $j = 1, \ldots, r$ is identical to the sum over $j = 0, \ldots, r-1$:

$$
\sum_{j=1}^{r} e^{-2\pi i s j/r} \, |a^{j} \bmod N\rangle = \sum_{j=0}^{r-1} e^{-2\pi i s j/r} \, |a^{j} \bmod N\rangle = \sqrt{r} \, |u_s\rangle.
$$

Therefore:

$$
U_a|u_s\rangle = e^{2\pi i s/r} \, |u_s\rangle.
$$

Note on the shift:
The shift $j = k + 1$ changes the exponent from $e^{-2\pi i s k/r}$ to $e^{-2\pi i s (j-1)/r}$. To restore the original form $e^{-2\pi i s j/r}$, we multiply by $e^{2\pi i s/r}$, which becomes the eigenvalue. This is the algebraic essence of the derivation.

Summary of the Derivation

$$
\begin{aligned}
U_a|u_s\rangle &= \frac{1}{\sqrt{r}}\sum_{k=0}^{r-1} e^{-2\pi i s k/r}\,|a^{k+1} \bmod N\rangle && \text{(definition)}\\
&= \frac{1}{\sqrt{r}}\sum_{j=0}^{r-1} e^{-2\pi i s (j-1)/r}\,|a^{j} \bmod N\rangle && \text{(index shift } j = k+1 \text{ modulo } r\text{)}\\
&= e^{2\pi i s/r}\cdot\frac{1}{\sqrt{r}}\sum_{j=0}^{r-1} e^{-2\pi i s j/r}\,|a^{j} \bmod N\rangle && \text{(factor out constant phase)}\\
&= e^{2\pi i s/r}\,|u_s\rangle && \text{(recognise } |u_s\rangle\text{)}.
\end{aligned}
$$

This explicitly shows that $|u_s\rangle$ is an eigenstate of U_a with eigenvalue $e^{2\pi i s/r}$. The key step is the cyclic shift of the index, which introduces the multiplicative phase factor.

How to prepare the target state

We cannot prepare a specific $|u_s\rangle$ because r and s are unknown.

However, the state $|1\rangle = |a^0 \bmod N\rangle$ is trivial to prepare. Observe:

$$
\frac{1}{\sqrt{r}}\sum_{s=0}^{r-1}|u_s\rangle = \frac{1}{r}\sum_{k=0}^{r-1}\left(\sum_{s=0}^{r-1} e^{-2\pi i s k/r}\right)|a^k \bmod N\rangle.
$$

The inner sum over s is r when $k = 0$, and 0 otherwise. Thus only $k = 0$ survives:

$$
\frac{1}{\sqrt{r}}\sum_{s=0}^{r-1}|u_s\rangle = |a^0 \bmod N\rangle = |1\rangle.
$$

Therefore, $|1\rangle$ is a uniform superposition of **all** eigenstates $|u_s\rangle$.

Why $|1\rangle$ is the Sum of All Eigenstates

Let us sum all r eigenstates with equal amplitude:

$$\frac{1}{\sqrt{r}}\sum_{s=0}^{r-1}|u_s\rangle = \frac{1}{\sqrt{r}}\sum_{s=0}^{r-1}\frac{1}{\sqrt{r}}\sum_{k=0}^{r-1}e^{-2\pi i\, sk/r}\,|a^k \bmod N\rangle$$

Swap the sums:

$$= \frac{1}{r}\sum_{k=0}^{r-1}\left(\sum_{s=0}^{r-1}e^{-2\pi i\, sk/r}\right)|a^k \bmod N\rangle$$

For a fixed k, the sum $\sum_{s=0}^{r-1} e^{-2\pi i\, sk/r}$ is:

- If $k = 0$: every term is $e^0 = 1$, so the sum is r.
- If $k \neq 0$: the sum of the r-th roots of unity is 0.

Therefore, only the $k = 0$ term survives:

$$\frac{1}{r}\sum_{k=0}^{r-1}\big(r\cdot\delta_{k,0}\big)|a^k \bmod N\rangle = |a^0 \bmod N\rangle = |1\rangle$$

$$\frac{1}{\sqrt{r}}\sum_{s=0}^{r-1}|u_s\rangle = |1\rangle$$

The easily prepared state $|1\rangle$ is a uniform superposition of **all** the eigenstates $|u_s\rangle$.

Why This Matters for Phase Estimation

- We cannot prepare a specific $|u_s\rangle$ because we do not know r or s.
- But we **can** prepare $|1\rangle$ trivially.

Phase Estimation on the Superposition $|1\rangle$

Recall $|1\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle$. We run QPE with m qubits in the first register and $|1\rangle$ in the second.

Step 1 – Controlled-U_a powers

For a single eigenstate $|u_s\rangle$ with eigenvalue $e^{2\pi i \phi_s}$, where $\phi_s = s/r$, the first register becomes:

$$\frac{1}{\sqrt{2^m}} \sum_{x=0}^{2^m-1} e^{2\pi i\, x\, \phi_s}\, |x\rangle.$$

By linearity, the full state after controlled operations is:

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \left(\frac{1}{\sqrt{2^m}} \sum_{x=0}^{2^m-1} e^{2\pi i\, x\, \phi_s}\, |x\rangle \right) \otimes |u_s\rangle.$$

Step 2 – Inverse Quantum Fourier Transform

The inverse QFT maps $\frac{1}{\sqrt{2^m}} \sum_x e^{2\pi i x \phi} |x\rangle$ to a state concentrated at the m-bit integer $2^m \phi$ (rounded). For $\phi = \phi_s$, this integer is an m-bit binary approximation of the phase:

$$\mathrm{QFT}^\dagger \left(\frac{1}{\sqrt{2^m}} \sum_{x=0}^{2^m-1} e^{2\pi i\, x\, \phi_s}\, |x\rangle \right) \approx |\, \mathrm{bin}_m(\phi_s)\, \rangle.$$

Applying QFT† to the first register therefore gives:

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |\, \mathrm{bin}_m(\phi_s)\, \rangle \otimes |u_s\rangle.$$

Step 3 – Measurement and Phase Recovery

Measuring the first register yields an m-bit binary string that directly encodes an approximation to ϕ_s for some random s. Interpreting the measurement outcome c as an integer, we have:

$$\frac{c}{2^m} \approx \phi_s = \frac{s}{r}.$$

The continued fraction algorithm then recovers the denominator r from the fraction $c/2^m$.

Recall that we Know c and 2^m but not s or r

The fraction $c/2^m$ is a known rational number that approximates the unknown fraction s/r.

Any rational number $x = c/2^m$ can be expanded as a continued fraction:

$$x = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots}}}.$$

Truncating this expansion at various depths gives a sequence of **convergents** p/q, which are the best rational approximations to x with small denominators.

Because $c/2^m \approx s/r$, the fraction s/r will appear (or be very close to) one of the convergents p/q. We simply check each convergent's denominator q by testing whether $a^q \equiv 1 \pmod{N}$.

Example ($N = 15$, $a = 7$, $r = 4$)

Possible phases: $\phi_s \in \{0, 1/4, 2/4, 3/4\}$.

- If $s = 1$: $\phi_1 = 1/4 = 0.25$. With $m = 8$, $c \approx 64$. $c/2^8 = 64/256 = 1/4$. Continued fraction gives denominator $4 \to r = 4$.
- If $s = 2$: $\phi_2 = 2/4 = 1/2$. $c \approx 128$. $128/256 = 1/2$. Convergent $1/2$ gives denominator 2. But $7^2 \bmod 15 = 4 \neq 1$, so we reject $r = 2$ and try another measurement.
- If $s = 3$: $\phi_3 = 3/4 = 0.75$. $c \approx 192$. $192/256 = 3/4$. Convergent $3/4$ gives denominator $4 \to r = 4$.

The two perspectives are equivalent

- **Period-finding view (Lecture 6c):** U_f creates entanglement between $|x\rangle$ and $|a^x \bmod N\rangle$; measuring the work register gives a periodic comb; QFT reveals the period spacing.
- **Phase-estimation view (Lecture 7):** Controlled powers of U_a kick back phases $e^{2\pi i s/r}$ onto control qubits; the inverse QFT reads out the binary fraction s/r.

Both views rely on the same controlled-$U_a^{2^j}$ gates. Phase estimation simply interprets the output differently – it transfers phase information to the control register instead of storing a product in a work register.

General Unitary $U = e^{i2\pi H}$

If $U = e^{i2\pi H}$ for some Hermitian H, then:

- Eigenvectors of U are the same as eigenvectors of H.
- Eigenvalues of U are $e^{i2\pi\lambda}$ where λ are eigenvalues of H.
- Phase estimation directly yields the eigenvalues λ of H.

This is crucial for **quantum simulation** – we can compute energies of physical systems.

Complexity and Circuit Implementation 🛠️

Gate Count

For n-bit precision:

- n Hadamard gates (initialization)
- Controlled-U^{2^j} for $j = 0, \ldots, n-1$
- Inverse QFT: $O(n^2)$ gates

If implementing U^{2^j} requires $O(\mathrm{poly}(j))$ gates (e.g., modular exponentiation in Shor), total cost

is $O(n^2)$ plus the cost of controlled-U powers.

Implementing Powers of U

The most challenging part is **efficiently implementing** U^{2^j}. For Shor's algorithm, modular exponentiation can be done with $O(n^3)$ gates using fast multiplication circuits.

For general U, we may need to **Trotterize** (in simulation) or rely on hardware that naturally implements U.

Circuit Depth

Phase estimation is relatively deep – it requires many sequential controlled operations. This is a challenge for near-term noisy devices, but it's the backbone of many quantum algorithms.

Iterative Phase Estimation 🔁

The Idea

Standard QPE uses m control qubits and an m-qubit inverse QFT. Can we do it with **fewer qubits**?

Yes! **Iterative (or semi-classical) phase estimation** uses **one** control qubit and reuses it multiple times, with classical feedback.

The Iterative Circuit

iterative phase estimation

1. Start with the **least significant bit**.

2. Use one control qubit, apply controlled-$U^{2^{m-1}}$, measure.
3. Based on the measurement, apply a classically controlled rotation to compensate the phase for the next bit.
4. Repeat for $m-2, \ldots, 0$.

Advantages

- **Qubit-efficient:** Only one control qubit + the eigenstate register.
- **Depth vs. width tradeoff:** More measurements and classical processing, but shallower quantum circuit.

Iterative phase estimation is often more practical for near-term devices.

PennyLane Demos 💻

1. Basic Phase Kickback with One Qubit

```
import pennylane as qml
import numpy as np

dev = qml.device('default.qubit', wires=2)

@qml.qnode(dev)
def phase_kickback(phi):
    # Target qubit in |1> (eigenstate of RZ)
    qml.PauliX(wires=1)

    # Control in |+>
    qml.Hadamard(wires=0)

    # Controlled-U: U = RZ(2*phi)
    qml.CRZ(2 * phi, wires=[0, 1])

    # Apply H to control
    qml.Hadamard(wires=0)

    return qml.probs(wires=0)

phi = np.pi / 2  # 90 degrees
probs = phase_kickback(phi)
print(f"P(0) = {probs[0]:.3f}, P(1) = {probs[1]:.3f}")
print(f"Expected: P(0) = cos^2(phi/2) = {np.cos(phi/2)**2:.3f}")
```

---

2. Two-Qubit Phase Estimation (Exact θ)

```
dev2 = qml.device('default.qubit', wires=3)  # 2 control + 1 target

@qml.qnode(dev2)
def two_qubit_pe(theta):
    # Prepare eigenstate |1> on target
    qml.PauliX(wires=2)

    # Superposition on controls
    for i in range(2):
        qml.Hadamard(wires=i)

    # Controlled-U^2 on first control (j=1)
    qml.ctrl(qml.RZ, control=1)(2 * 2 * np.pi * theta, wires=2)

    # Controlled-U on second control (j=0)
    qml.ctrl(qml.RZ, control=0)(2 * 1 * np.pi * theta, wires=2)

    # Inverse QFT on 2 qubits
    qml.adjoint(qml.QFT)(wires=[0,1])

    return qml.probs(wires=[0,1])

# Test theta = 0.75 = 3/4 = 0.11_2
theta = 0.75
probs = two_qubit_pe(theta)
print("Probabilities for |00>,|01>,|10>,|11>:")
print(probs.round(3))
# Should peak at |11>
```

3. General Phase Estimation with PennyLane's Template

```
n_est = 4  # estimation qubits
dev_pe = qml.device('default.qubit', wires=n_est + 1)

@qml.qnode(dev_pe)
def qpe(theta):
    # Prepare eigenstate |1>
    qml.PauliX(wires=n_est)

    # Phase estimation circuit
    qml.QuantumPhaseEstimation(
        unitary=qml.RZ(2 * np.pi * theta, wires=n_est),
        estimation_wires=range(n_est)
    )

    return qml.probs(wires=range(n_est))

theta = 0.3125  # = 5/16 = 0.0101_2
probs = qpe(theta)
most_likely = np.argmax(probs)
print(f"Most likely outcome: {most_likely} (binary {format(most_likely, '04b')})")
print(f"Estimated theta: {most_likely / 2**n_est:.4f} (true = {theta:.4f})")
```

Summary and Key Takeaways 🎯

What We Learned

1. **Phase kickback** is the fundamental mechanism: a controlled-U with target in an eigenstate transfers the eigenvalue phase to the control qubit.
2. **Powers of** U shift the binary digits of the phase, allowing us to read off bits sequentially.
3. **Quantum Phase Estimation (QPE)** combines these ideas with the inverse QFT to estimate an unknown phase θ to n-bit precision.
4. **Applications** include Shor's algorithm (period finding = eigenvalue estimation),

quantum chemistry, and more.

5. **Iterative QPE** reduces qubit requirements at the cost of classical feedback.

Complexity Summary

Algorithm Step	Gate Count
Hadamard initialization	$O(n)$
Controlled-U^{2^j}	Depends on U
Inverse QFT	$O(n^2)$
Total	$O(n^2)$ + cost of U

Exercises 📝

1. **Phase Kickback Verification**
   For $U = Z$ (Pauli-Z) and target $|1\rangle$, compute the state after controlled-Z with control initially in $|+\rangle$. Show that the control becomes $|-\rangle$. What is the phase kickback?
2. **Binary Fraction Practice**
   Express $\theta = 0.625$ as a binary fraction. What are the phases for U, U^2, U^4?
3. **Two-Qubit QPE by Hand**
   For $\theta = 0.625$ ($= 5/8 = 0.101_2$), simulate the 2-qubit QPE circuit by hand and verify that the final state peaks at $|10\rangle$ (since $2^2 \times 0.625 = 2.5$, the closest integer is 2, which is 10_2).
4. **PennyLane Simulation**
   Modify the general QPE code to estimate $\theta = 0.123$. How many estimation qubits are needed to get an accuracy of ± 0.01? What is the success probability?
5. **Iterative QPE**
   Implement iterative phase estimation for $n = 3$ bits in PennyLane. Compare the circuit depth with the standard QPE.
6. **Shor's Eigenstate Sum**

Prove that $\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$ for the modular exponentiation unitary. Then explain why this allows us to use $|1\rangle$ as the input to phase estimation.

Next Lecture Preview ⏩

Grover's Search Algorithm

- Unstructured search problem
- Amplitude amplification
- Geometric interpretation
- Optimality and applications

AI Tool Demo 🤖

This lecture's content and code were prepared with the assistance of AI (DeepSeek). The explanations, examples, and PennyLane code were generated iteratively based on the course plan.

Remember: AI tools are powerful assistants, but always verify the mathematics, run the code, and understand the underlying concepts. Quantum computing requires both intuition and rigorous understanding – the AI can help build intuition, but you must own the rigor.

"Phase estimation is the Swiss Army knife of quantum algorithms – once you can estimate eigenvalues, you can factor numbers, simulate physics, and solve linear systems." 🚀